# Data Visualization

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## Accessibility Notice

This document is also available in HTML format at:
`https://aholdengouveia.name/IntroData/labs/datavisualizations.html`

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

## Learning Outcomes

By the end of this lab, you will be able to:

- Evaluate data visualizations using published guidelines and a checklist
- Explain what makes a visualization good, bad, or misleading
- Make a visualization more accessible, with alt text and a narration
- Create visualizations of a data set, and explain your choices

## Lab Instructions

Examples of good Data Visualizations

- Information is Beautiful `https://informationisbeautiful.net/`
- Storytelling with data `https://www.storytellingwithdata.com/blog`

Examples of bad Data Visualizations

- WTF Visualizations: Visualizations that make no sense `https://viz.wtf/`
- r/dataisugly `https://www.reddit.com/r/dataisugly/`

## Evaluate visualizations

Using the guidelines from this website `https://ixdf.org/literature/article/guidelines-for-good-visual-information-representations` and the materials from this week, answer the following questions. For questions 1 to 3, use the table and checklist (DataVizChecklist) to break down your answer and make sure it hits each of the guidelines. The checklist is linked here as a Word document Link: `https://www.aholdengouveia.name/IntroData/labs/DataVizChecklist.docx` or as an OpenDocument file Link: `https://www.aholdengouveia.name/IntroData/labs/DataVizChecklist.odt`

1. Your favorite visualization
   (a) Include a screenshot of the visualization you picked and its URL.
   (b) Why is it your favorite?
   (c) Fill out the table and checklist for it.
2. Your least favorite or most aggravating visualization
   (a) Include a screenshot of the visualization you picked and its URL.
   (b) Why is it your least favorite?
   (c) Fill out the table and checklist for it.
3. The visualization you found most humorous. This can come from any of the example websites.
   (a) Include a screenshot of the visualization you picked and its URL.
   (b) Why did you find it the most humorous?
   (c) Fill out the table and checklist for it.
4. Accessibility: from the good examples, pick one you like that you haven't already evaluated. We're going to look at the accessibility of it.
   (a) Include a screenshot of the visualization you picked and its URL.
   (b) Name 3 ways you can make this visualization more accessible.
   (c) Write an appropriate alt text for it.
   (d) Create a narration of the visualization. Your narration can be written or a video. One easy way to make a video is to create a screen capture video using something like OBS, upload it to a site like YouTube, and make it unlisted so you can just share a link.

5. Go to `https://www.economist.com/graphic-detail`, do you think in general, this link belongs in the good or bad category? You don't need to evaluate every visualization, just look through the first page or two, and explain in a paragraph which one you think this link belongs in, why, and include screenshots of the visualizations you picked that support your opinion.

### Create your own Visualizations

1. Going back to the Halloween candy data, pick 1 thing that you think would be better illustrated using a visualization. Create your visualization using whatever software you like, this could be an online chart maker, or a spreadsheet or anything.
2. Explain why you picked that 1 thing, and explain how you decided which visualization to use and why.
3. Go back to the data you collected for the Data Collection lab in Week 2, and pick 2 things that you think would be better illustrated using a visualization. Create your visualization using whatever software you like, this could be an online chart maker, or a spreadsheet or anything.
4. Explain why you picked those 2 things from your data, and explain how you decided which visualization to use and why.

## Deliverables

1. A document with answers to each of the above questions evaluating other people's visualizations. Make sure to include a screenshot and the URL of each visualization you picked, and fill out the linked table and checklist for each of the visualizations in questions 1 to 3.
2. A document for your visualizations, the document should include screenshots (do not just upload multiple images to Brightspace) of the 3 visualizations that you created. Each screenshot should have the explanation of why below it.

## Resources

References, a video, a PowerPoint and some notes are available at my website Link: `https://www.aholdengouveia.name/IntroData/datavis.html`